Weighted Projective Spaces

Davide Borra (264683)
University of Trento

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Abstract

This seminar is held as final exam for the course "Toric Geometry", with the supervision of Prof. E. Postinghel and Prof. L.E. Solá Conde. It is an application of the theory we have developed in the course to the simple but interesesting example of weighted projective spaces.

1 Classical Construction

When we work with $\mathbb{P}^n$, we define subvarieties by considering homogeneous polynomials in $\mathbb{C}[x_0, \ldots, x_n]$. This is the polynomial ring in $n+1$ variables, and is a graded ring where each monomial x_i has degree 1. One might wonder if it is possible to have a similar variety where these monomials have different degrees, and indeed such a construction exists and defines a toric variety: the weighted projective space.

Consider $q_1, \ldots, q_n$ positive integers with $\gcd(q_0, \ldots, q_n) = 1$. Then we can define the *weighted projective space*

$$\mathbb{P}(q_0, \ldots, q_n) := (\mathbb{C}^{n+1} \setminus \{0\})/\sim,$$

where $\sim$ is the equivalence relation given by

$$(a_0, \ldots, a_n) \sim (b_0, \ldots, b_n) \iff \exists \lambda \in \mathbb{C}^* : a_i = \lambda^{q_i} b_i \ \forall i = 0, \ldots, n.$$

This is nothing else but usual construction of $\mathbb{P}^n$ with some weights q_i added, and indeed $\mathbb{P}(1, \ldots, 1) = \mathbb{P}^n$. It is quite trivial to notice that if we picked the q_i's with a common factor d, the construction would've been the same as before, simply by choosing a different λ.

2 Toric construction

An interesting fact about weighted projective spaces is that they are toric varieties. If indeed we pick the q_i's as before, we can define $N := \mathbb{Z}^{n+1}/\mathbb{Z}(q_0, \ldots, q_n)$, and let u_i $(i = 0, \ldots, n)$ be the image of the i-th element of the canonical basis of $\mathbb{Z}^{n+1}$ through the quotient map. Then the relation $\sum_i q_i u_i = 0$ holds by construction. Consider now Σ to be the fan

$$\Sigma := \left\{ \operatorname{Cone} U \;\middle|\; U \subsetneq \{u_0, \ldots, u_n\} \right\},$$

then the variety X_Σ will be exactly $\mathbb{P}(q_0, \ldots, q_n)$.

While describing the classical construction, we noticed that it is non-restrictive to assume the q_i's to be coprime: now we show that we may ask even more. A weighted projective space is said to be *well-formed* if no n weights share a common factor. We now show that every weighted projective space is isomorphic to a well-formed one. If we call $d_i := \gcd(q_0, \dots, q_{i-1}, q_{i+q}, \dots, q_n)$: then we can write

$$u_i q_i = -\sum_{j \neq i} d_i q_j^{(i)} u_j = -d_i \sum_{j \neq i} q_j^{(i)} u_j$$

and, since $\gcd(q_0, d_i) = 1$, then d_i must divide u_i, thus we can replace it as a generator of the associated ray with $u_i' = \frac{1}{d_i} u_0$. If we proceed this way for every i, we easily get that the relation

$$\sum_{i=0}^{n} q_i' u_i = 0$$

is satisfied if we set $q_i' := q_i / \prod_{j \neq i} d_j$ (this division makes sense as all the d_i's are pairwise coprime by construction), thus concluding the two spaces are indeed isomorphic. This is as we changed the generators of the rays, but the fan we obtain is still the same.

3 Divisors in weighted projective spaces

3.1 Divisor Class Group

By the previous toric construction, we know that the fan Σ describing $\mathbb{P}(q_0, \dots, q_n)$ has exactly $n+1$ rays, hence

$$\mathrm{Div}^T(\mathbb{P}(q_0, \dots, q_n)) \cong \mathbb{Z}^{n+1}.$$

Moreover, the previous construction gives a short exact sequence

$$0 \longrightarrow M \xhookrightarrow{\alpha} \mathbb{Z}^{n+1} \xrightarrow{\beta} \mathbb{Z} \longrightarrow 0$$

where α is the inclusion of $M = \{(a_0, \dots, a_n) \in \mathbb{Z}^{n+1} \mid \sum_i a_i q_i = 0\}$ in $\mathbb{Z}^{n+1}$ and β is the map $(a_0, \dots, a_m) \mapsto \sum_i a_i q_i$. The exactness at M and at $\mathbb{Z}^{n+1}$ is trivial by construction, while the surjectivity of β is ensured by the fact that the $\gcd(q_1, \dots, q_n) = 1$. Since the rays of Σ span M, we may combine this with [CLS11, Theorem 4.1.3] and conclude that $\mathrm{Cl}(\mathbb{P}(q_0, \dots, q_n)) \cong \mathbb{Z}$.

We can give an explicit description of the generator of the divisor class group. We already know that

$$\mathrm{Div}^T(\mathbb{P}(q_0, \dots, q_n)) = \bigoplus_{\rho \in \Sigma(1)} \mathbb{Z} D_\rho,$$

where the isomorphism used above is the one mapping each principal divisor to an element of the canonical basis (write $\Sigma(1) = \{\rho_0, \dots, \rho_n\}$). Now, the generator of $\mathrm{Cl}(\mathbb{P}(q_0, \dots, q_n))$ the class of any Weil divisor sent to 1: since the q_i's are coprime, there exist coefficients $b_i \in \mathbb{Z}$ such that $\sum_i b_i q_i = 1$: then

$$\mathrm{Cl}(\mathbb{P}(q_0, \dots, q_n)) = \mathbb{Z}[D], \qquad D = \sum_{i=0}^{n} b_i D_{\rho_i}.$$

3.2 Picard Group

We simply need to determine for which $k \in \mathbb{Z}$ the divisor $kD = \sum_{i=0}^{n} k b_i D_{\rho_i}$ is Cartier. By [CLS11, Theorem 4.2.8], this is equivalent to determining for which k's for all maximal cones $\sigma \in \Sigma$ there exists an element $m_\sigma \in M$ satisfying $\langle m_\sigma, u_i \rangle = -k b_i$ for all i's such that $\rho_i \prec \sigma$. By construction, we know that the maximal cones of Σ are of the form $\sigma_j = \mathrm{Cone}(u_0, \dots, u_{j-1}, u_{j+1}, \dots, u_j)$. Moreover, by the previous identification of M with a subset of

$\mathbb{Z}^{n+1}$, also may write $m_{\sigma_j} = m^{(j)} = (m_0^{(j)}, \dots, m_n^{(j)})$, thus obtaining $\langle m^{(j)}, u_i \rangle = m_i^{(j)}$, so that the condition restates as

$$\forall j = 0, \dots, n, \ \exists m^{(j)} \in M \text{ such that } m_i^{(j)} = -kb_i \forall i \neq j.$$

Using the description of M above, for $m^{(j)} = (-kb_0, \dots, -kb_{j-1}, m_j^{(j)}, -kb_{j+1}, \dots, -kb_n)$, we have

$$\begin{aligned}
m^{(j)} \in M &\iff \sum_{i=0}^{n} m_i^{(j)} = 0 \\
&\iff -k \sum_{\substack{i=0 \\ i \neq j}}^{n} b_i q_i = -m_j^{(j)} q_j \\
&\iff k(1 - b_j q_j) = m_j^{(j)} q_j && \text{by choice of the } b_i\text{'s} \\
&\iff q_j \mid k(1 - b_j q_j) \\
&\iff q_j \mid k && \text{as } \gcd(1 - b_j q_j, q_j) = 1.
\end{aligned}$$

This proves that kD is Cartier if and only if $k = k' \operatorname{lcm}(q_0, \dots, q_n)$ for some $k' \in \mathbb{Z}$, and therefore $\operatorname{Pic}(\mathbb{P}(q_0, \dots, q_n)) = \mathbb{Z}[\operatorname{lcm}(q_0, \dots, q_n)D]$, and

$$\operatorname{Cl}(\mathbb{P}(q_0, \dots, q_n)) / \operatorname{Pic}(\mathbb{P}(q_0, \dots, q_n)) \cong \mathbb{Z} / \operatorname{lcm}(q_0, \dots, q_n)\mathbb{Z},$$

hence every weighted projective space is $\mathbb{Q}$-factorial. In particular, we have

$$m_i^{(j)} = \begin{cases} -kb_i & i \neq j \\ \frac{k}{q_j}(1 - b_j q_j) & i = j \end{cases}$$

3.3 Base point free and ample divisors

Consider now any Cartier divisor — up to linear equivalence we may assume it to be hkD for $h \in \mathbb{Z}$, $k = \operatorname{lcm}(q_0, \dots, q_n)$ and where D is as defined above. Then we start by computing the polytope associated to hkD, which is

$$\begin{aligned}
P(khD) &= \{m \in M_{\mathbb{R}} \mid \langle m, u_\rho \rangle \geq -a_\rho \ \forall \rho \in \Sigma(1)\} \\
&\cong \{m \in M_{\mathbb{R}} \subseteq \mathbb{R}^{n+1} \mid m_i \geq -hkb_i \ \forall i = 0, \dots, n\}
\end{aligned}$$

where in the second description we are using the identification of M with the kernel of the linear map $\begin{pmatrix} q_0 & \cdots & q_n \end{pmatrix} : \mathbb{R}^{n+1} \to \mathbb{R}$, as above. If we proceed as before, we compute the coefficients $m^{(j)}$ associated to hkD, given by

$$m_i^{(j)} = \begin{cases} -hkb_i & i \neq j \\ \frac{hk}{q_j}(1 - b_j q_j) & i = j \end{cases}$$

Then, by [CLS11, Proposition 6.1.1], the divisor is base point free if and only if $m^{(j)} \in P(khD)$ for every $j = 0, \dots, n$: this condition simplifies to

$$\begin{cases} -hkb_i \geq -hkb_i & \forall i \neq j \\ \frac{h\cancel{k}}{q_j}(1 - b_j q_j) \geq -h\cancel{k} b_j \end{cases} \iff h(1 - b_j q_j) \geq -hb_j q_j \iff h \geq 0$$

hence hkD is base point free whenever $h \geq 0$.

The discussion about ampleness is quite similar: by combining [CLS11, Lemma 6.1.13, Theorem 6.1.14], we obtain that the divisor is ample if and only if for every maximal cone σ, it holds $\langle m_\sigma, u_\rho \rangle > -a_\rho \ \ forall \rho \in \Sigma(1) \smallsetminus \sigma(1)$. In our notation, this means that $m_j^{(j)} > -hkb_j$ for every $j = 0, \dots, n$: as before, this reduces to

$$\frac{h\cancel{k}}{q_j}(1 - b_j q_j) > -h\cancel{k}b_j \qquad \Longleftrightarrow \qquad h(1 - b_j q_j) > -hb_j q_j \qquad \Longleftrightarrow \qquad h > 0$$

and therefore the divisor is ample if and only if $h > 0$.

4 A more hands-on example: $\mathbb{P}(1, \dots, 1, q)$

In this section, we discuss a more hands-on example, where $q_0, \dots, q_{n-1} = 1$ and $q_n = q$, for $q > 1$. In this case, a more explicit description of the lattice is possible: indeed $\mathbb{Z}^{n+1}/(1, \dots, 1, q)\mathbb{Z}$ is isomorphic to $\mathbb{Z}^n$, and the images of the vectors in the canonical bases are

$$\begin{cases} e_i \mapsto u_i & i > 0 \\ e_0 \mapsto u_0 = -\sum_{i=1}^{n-1} u_i - qu_n, \end{cases}$$

where $u_1, \dots, u_n$ is the canonical basis of $\mathbb{Z}^n$. This gives a way of interpreting the rays more explicitly, and enables us to describe a polytope associated to $\mathbb{P}(1, \dots, 1, q)$: since the rays are the spans with positive coefficients of the u_i's, then a possible choice of a polytope whose normal fan is that of $\mathbb{P}(1, \dots, 1, q)$ is

$$P = \left\{ (x_1, \dots, x_n) \in \mathbb{R}^n \;\middle|\; x_i \geq 0 \ \forall i, \ \sum_{i=1}^{n-1} x_i + qx_n \leq q \right\}.$$

It is quite easy to notice that this is no other than

$$P \cap M = \big((q\Delta_{n-1} \cap \mathbb{Z}^{n-1}) \times \{0\}\big) \cup \{(0, \dots, 0, 1)\},$$

hence the induced embedding will be given by

$$\begin{array}{ccc} (\mathbb{C}^*)^n & \longrightarrow & \mathbb{P}^N \\ t & \longmapsto & [\cdots : t^\alpha : \cdots : t_n] \quad \alpha \in \mathbb{N}^{n-1}, |\alpha| \leq q. \end{array}$$

Where $N = \binom{n+q-1}{q}$ is the size of P. At first sight, this is not that meaningful, but if we focus for a moment only on the first $n-1$ components, we get

$$\begin{array}{ccc} (\mathbb{C}^*)^{n-1} & \longrightarrow & \mathbb{P}^{N-1} \\ t & \longmapsto & [\cdots : t^\alpha : \cdots] \quad \alpha \in \mathbb{N}^{n-1}, |\alpha| \leq q \end{array}$$

which can be completed via a simple homogenization ($\alpha \leftrightarrow \alpha' = (q - |\alpha|, \alpha)$) to

$$\begin{array}{ccc} \mathbb{P}^{n-1} & \longrightarrow & \mathbb{P}^{N-1} \\ x & \longmapsto & [\cdots : x^{\alpha'} : \cdots] \quad \alpha' \in \mathbb{N}^n, |\alpha'| = q \end{array}$$

and this is clearly the q-Veronese embedding of $\mathbb{P}^{n-1}$. Thus, we obtain that the whole variety is a cone on the q-Veronese (embedded in the hyperplane given by the vanishing of the last coordinate of $\mathbb{P}^N$) with vertex $[0 : \cdots : 0 : 1]$.

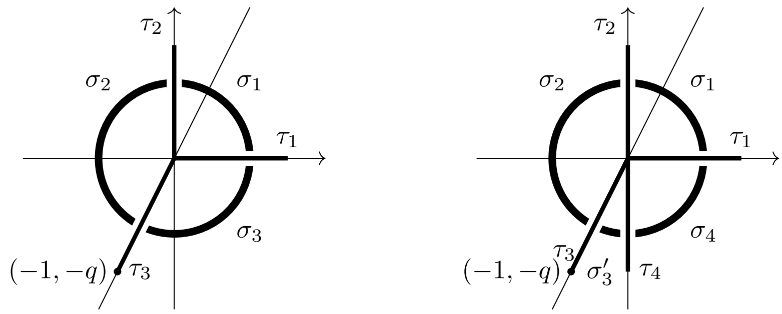
4.1 Singularities and desingularization

Recall that the maximal cones in the fan describing $\mathbb{P}(1,\dots,1,q)$ are the one generated by all the u_i's but one. If we look at the determinant of the matrix

$$A_j = \left(\begin{array}{ccc|c} \cdots & \underset{i\neq 0,j}{u_i} & \cdots & u_0 \end{array} \right),$$

since the u_i's are a basis of $\mathbb{Z}^n$ and $u_0 = (-1,\dots,-1,-q)$, we get that the determinant is always ± 1, except when $j = n$: in this case the determinant is $\pm q$, the sign depending on the size of the matrix. This tells us we have only one singular point, which must be the vertex of the cone: $[0 : \cdots : 0 : 1]$. One interesting thing we might do is try to remove this singularity by blowing up $\mathbb{P}(1,\dots,1,q)$ at $[0 : \cdots : 0 : 1]$. Since we are working with a toric variety, this is simply a matter of performing a star subdivision of the cone generated by $u_0,\dots,u_{n-1}$: this means adding the ray containing $u_0 + \cdots + u_{n-1} = -qu_n$, whose generator is $-u_n$. A simple computation shows the new variety we obtain is indeed smooth.

The procedure we described in the general case can be seen more easily if we look at the two-dimensional case. In this case, the blowing up of $\mathbb{P}(1,\dots,1,q)$ at the singular poing is exactly the Hirzebruch surface $\mathbb{F}_q$, which is indeed smooth.



(a) Before the star subdivision

(b) After the star subdivision

Figure 1

4.2 Linear systems

In this last two-dimensional case, we can easily compute linear systems: pick an ample divisor hkD (with $h > 0$, k and D as in the general case) and write $X = \mathbb{P}(1,1,q)$. Then we get a map to the projective space

$$\varphi : X \dashrightarrow \mathbb{P}(\Gamma(X, \mathscr{O}_X(hkD))^\vee)$$

which is indeed a morphism, as we assumed the line bundle to be base point free. By [CLS11, §6.1], we know that $\{x^m \mid m \in P_{hkD}\}$ is a basis for $\Gamma(X, \mathscr{O}_X(hkD))$, hence, if we call $s^m := (x^m)^\vee$, we may write

$$\begin{array}{rccc} \varphi : & X & \longrightarrow & \mathbb{P}^N \\ & x & \longmapsto & [s^{m^0}(x) : \cdots : s^{m^N}(x)] \end{array}$$

where $P(hkD) = \{m^0, \cdots, m^N\}$. It only remains to describe explicitly $P(hkD)$: start by recalling what we did in the general case,

$$\begin{aligned} P(khD) &= \{m \in M_{\mathbb{R}} \mid \langle m, u_\rho \rangle \geq -a_\rho \ \forall \rho \in \Sigma(1)\} \\ &\cong \{m \in M_{\mathbb{R}} \subseteq \mathbb{R}^{n+1} \mid m_i \geq -hkb_i \ \forall i = 0, \ldots, n\}, \end{aligned}$$

where in the second row, $M_{\mathbb{R}}$ is interpreted as the kernel of the map $m \mapsto \langle (1, 1, q), m \rangle$. In this case we have $k = q$ and we can choose $b_0 = 1$, $b_2 = b_3 = 0$, then

$$\begin{aligned} P(khD) &= \{(m_1, m_2, m_3) \in \mathbb{R}^3 \mid m_1 \geq -hk, m_2 \geq 0, m3 \geq 0, m_1 + m_2 + qm_3 = 0\} \\ &\cong \{(x, y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0, x + qy \leq hq\}. \end{aligned}$$

In case $h = 1$, this can be easily drawn: in this example we can therefore have an explicit description of φ.

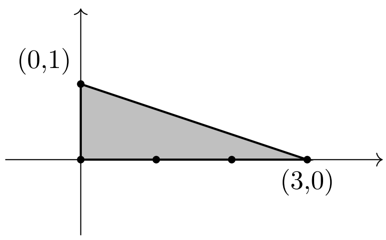


Figure 2: The polytope of the divisor hkD when $(k =)q = 3$ and $h = 1$

$$\begin{array}{rccc} \varphi: & X & \longrightarrow & \mathbb{P}^4 \\ & p & \longmapsto & [1 : x^\vee(p) : (x^2)^\vee(p) : (x^3)^\vee(p) : y^\vee(p)]. \end{array}$$

More in general, if we change q but keep $h = 1$, we will have

$$p \longmapsto [1 : x^\vee(p) : \cdots : (x^q)^\vee(p) : y^\vee(p)].$$

References

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[Mil02] Reid Miles. *Graded rings and varieties in weighted projective space*. 2002. URL: https://mreid.warwick.ac.uk/surf/more/grad.pdf.